

Quantum Gradient Time Crystal Dilation: Classical Temporal-Opportunity Kernels and a Conditional Newtonian Calibration

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Abstract

Quantum Gradient Time Crystal Dilation (QGTCD) is the historical name of a documented proposal in which otherwise symmetric microscopic motion becomes directionally biased when neighboring states carry unequal temporal opportunity. This paper asks the narrower question that can presently be answered: what follows when that proposal is represented by explicit classical stochastic kernels? Three finite-step kernels are nonidentical but have, under matched smooth-lattice assumptions, the same leading drift-diffusion generator. On a compatible periodic lattice, the endpoint kernel's exact stationary measures converge to the invariant density of that limiting diffusion. These results identify a continuum model class while showing that the historical narrative does not select a unique microscopic rule. Directly interpreting its overdamped drift as universal free fall fails because the response scales with probe diffusivity. A separate position-velocity calibration removes that dependence only by importing $u = -\Phi/c^2$ and postulating $a = c^2 \text{ grad } u$; Newtonian acceleration then follows by substitution, not by derivation. Additive velocity noise without dissipation increases kinetic variance and supplies no equilibrium model. Setting $c_* = c$ gives leading weak-field clock compatibility only after importing the standard clock map. The paper therefore proves conditional properties of a classical model family, defines an internal preparation-dependence discriminator, and states a preregistration-ready test structure. It does not derive a gravitational source law, equivalence principle, spacetime metric, light propagation, backreaction, quantum dynamics, or a prediction beyond general relativity. The current result is a bounded classical stochastic-model program and an explicit map of what a future gravity theory would still have to supply.

probabilistic gravity model; time dilation; Markov kernel

Keywords: QGTCD; biased random walk; temporal measure; drift diffusion;

Scope and evidence boundary

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Claim contract

This is a source-traced hypothesis and mathematical-model paper. It makes three different kinds of statement:

1. **Documentary statements** report what the dated QGTCD sources contain.
2. **Model theorems** establish properties of the transition kernel defined here, under explicit assumptions.
3. **Physical hypotheses** ask whether a field and kernel of this kind describe gravity in nature.

Only the second class is proved in this draft. A classical stochastic process does not become quantum merely because its outcomes are probabilistic. A drift coefficient or conditional mean displacement rate does not

become gravitational acceleration merely because it points toward a source. An inverse-square limit does not by itself supply relativistic covariance, light bending, gravitational radiation, or the equivalence principle. These boundaries are part of the result rather than disclaimers added after it.

Scope freeze

This preprint deliberately closes at a bounded classical endpoint: source-traced temporal-opportunity kernels, their controlled continuum limit, the failure of a direct overdamped free-fall interpretation, and a conditional Newtonian position-velocity calibration. A dynamical gravity theory, a microscopic or quantum completion, and a quantitative residual beyond general relativity are separate future research products. Their absence is not an unfinished promise inside this paper and is not repaired by expanding the interpretation of the present equations.

Result-status map

Statement	Status in this paper	What the status does not imply
Each defined positive finite-neighbor kernel is normalized	Proved under its stated assumptions	That nature uses any one kernel
The three kernels share a leading drift-diffusion generator	Proved by matched local-moment expansion on smooth Cartesian lattices; the supplied numerical regression is one-dimensional	Finite-step identity, arbitrary-graph convergence, or microscopic uniqueness
The endpoint chain converges weakly on a periodic compact domain	Proved conditionally	Convergence on arbitrary boundaries or geometries
Its discrete stationary measures converge to $p_{\infty} \propto \exp(2\beta u)$	Proved on compatible periodic lattices	Thermodynamic equilibrium or a gravitational energy law
A fixed-time response slope $s_D=1$ versus $s_D=0$ separates the overdamped and inertial implementations	Internal model discriminator using the same displacement observable	Evidence for QGTCD over standard gravity
$a = -\text{grad } \Phi$ follows from the inertial calibration	Conditional identity by substitution	A derivation of gravity, G , sourcing, or universality
Unequal temporal opportunity causes gravity in nature	Physical hypothesis	Established by the mathematical model
The model is quantum or realizes a physical time crystal	Not established	Implied by the historical name

The table is the reading contract for the remainder of the paper. Results may move to a stronger status only through a new derivation, independent review, or empirical evidence; interpretation alone cannot promote them.

1. Documentary origin of the mechanism

The first identified public source in the genealogy that later became QGTCD is a0253z.md, committed on 26 July 2022. It proposed that mass be treated as time-crystal-like, associated greater mass density with a spatial variation in time, and described motion toward the region of greater density. A 30 July 2022 revision made the stochastic atom explicit: in open space a particle's directional motion was described as random, whereas near mass the direction toward mass receives greater odds because additional temporal measure is treated as additional available spacetime measure. The same source proposed the descriptive phrase “quantum probability distortion from mass frequency” [1,2].

The exact name has its own genealogy. A 28 November 2022 changelog records that “Quantum Gradient Time Crystal Dilation” followed the earlier name “Quantum Gradient Time Dilation.” The public mechanism can therefore be dated to July 2022 without pretending that every later term or equation appeared on the first date [3].

In 2024, three public essays restated the mechanism as a quadrant or random-walk argument: equal temporal opportunity yields no net direction; unequal temporal opportunity changes directional odds; many microscopic deviations yield a curved mean trajectory. A two-postulate note dated 30 March 2024 gives the same core in compact form [4-7]. These texts establish a documented research proposal. They do not establish that the proposal is physically correct.

1.1 The November 2021 fourth-branch boundary

Three recovered authored notes precede the public probability mechanism. On 11 November 2021, a0033z.md records Micah’s own “equation for quantum gravity.” On 13 November, a0189z.md calls the physics material a “new 4th section.” On 20 November, a0044z.md repeatedly describes a new quantum-gravity equation and theory connected to the neuroscience program. A later Git state publicly fixes the sentence “Gravity is gradient time dilation at the particle & atomic scale” by 16 June 2022.

These records establish an authored branch boundary and explain the transition from SAN to a fourth physics program. They do **not** contain the July 2022 probability-bias atom, the later QGTCD name, the endpoint kernel, or the equations proved here. The distinction is deliberate:

Date layer	Supported statement	Statement not backdated into it
November 2021 authored notes	A new quantum-gravity branch existed in Micah’s private record	Public priority, QGTCD naming, or a normalized stochastic kernel
June 2022 Git fixation	Gradient time dilation was publicly attached to gravity at particle / atomic scale	The explicit directional odds mechanism
July 2022 public revisions	Unequal temporal opportunity changes directional odds toward mass	Any unique 2026 kernel or continuum theorem
November 2022 onward	The QGTCD name and larger equation family developed	Retrospective proof that all later equations were present in 2021-22

This genealogy is evidence about development and priority boundaries, not evidence that the physics is correct.

1.2 Terminology guard: “time crystal”

Time-crystalline order has a technical meaning in quantum many-body physics. For example, experimentally realized discrete time crystals exhibit ordered subharmonic response under driven dynamics. That literature establishes that time-crystalline phases exist; it does not establish that ordinary mass is a time crystal or that time-crystalline order causes gravity. In this paper, “time crystal” is retained in the historical name QGTCD. The formal model uses the neutral term **temporal-opportunity field** so the mathematics does not depend on an unproved identity between mass and condensed-matter time-crystalline order [8].

1.3 Time dilation is an observation, not the proposed cause

Gravitational redshift is experimentally established, including across a millimetre-scale atomic sample. Relativistic and quantum-clock research also shows that clock behavior in quantum states requires careful

definitions. Such results constrain any account of gravity; they do not independently validate QGTCD’s proposed causal direction from temporal measure to motion [9-11]. The question here is narrower: can the 2022 probability-distortion narrative be written as a coherent normalized stochastic model, and what does that model actually imply?

1.4 Sources cited at the time

The July 2022 note did not present itself in an intellectual vacuum. It linked time-crystal experiments and reporting, quantum-time-dilation work, gravitational-redshift reporting, and an author-identified book that might contain a similar conclusion. Those contemporaneous routes should remain visible in the genealogy.

Source class cited in a0253z.md	Role in the 2022 note	Boundary for this paper
Discrete time-crystal experiments and reporting	Inspiration and terminology context	Does not establish that ordinary mass is time-crystalline order or causes gravity
Quantum-clock and quantum-time-dilation papers	Prior research context explicitly acknowledged by Micah	Does not contain the QGTCD endpoint probability-bias kernel
Gravitational-redshift and clock reporting	Empirical context for spatially varying clock rates	Tests time dilation, not QGTCD’s causal mechanism
A possibly similar gravity book, raised as a question in the note	Prospective comparator identified by Micah	Requires a separate content / date audit before any priority disposition

This table narrows the priority claim rather than weakening it: the prior context is named, while the added directional temporal-opportunity mechanism is isolated for exact comparison.

QGTCD should also not be conflated with the established field called **stochastic gravity**, which extends semiclassical gravity through Einstein-Langevin dynamics and a quantum stress-energy noise kernel. The model below is a classical position-state Markov formalization of a different historical proposal [15].

1.5 Equation genealogy and the non-backdating rule

The QGTCD archive contains several equation families that answer different questions and do not yet form one derived theory. The present model paper keeps them visible without making later mathematics appear earlier or importing unsupported equations into the theorem set.

Layer	Equation or mathematical atom	Current role
2021 branch notes	An authored quantum-gravity equation and gradient-time-dilation language	Chronology; the exact later kernel is absent
July 2022 mechanism	Greater temporal opportunity changes directional probability	Historical modeling constraint, not a unique formula
2023 energy / frequency family	$m\tilde{c}^2 = (\hbar/N) f$, $E = (\hbar/N) f$, and related forms	Preserved conjectures; N remains undefined and one rearrangement is dimensionally contradicted
2023 modified wave equation	$\hbar \partial_t \psi = i \hbar \bar{N}(r,t) \partial_t \psi$	Unresolved quantum hypothesis; no action, covariance rule, or norm proof
March 2024 postulates	Uniform deviations cancel; a time-density gradient biases the mean path	Compact public statement of the stochastic atom
2026 endpoint model	$P_E(x,y) \propto \exp(\beta u(y))$	One transparent endpoint realization with proved conditional properties
2026 kernel-comparison model	Endpoint, centered-linear, and Metropolis kernels	Demonstrates finite-step non-uniqueness and a shared continuum limit under matched assumptions

The complete preservation-first disposition remains in LEGACY-EQUATION-AUDIT.md. No red or yellow legacy equation is silently promoted by the new continuum analysis.

2. Minimal state-graph model

Begin with an ordinary symmetric choice. A particle at one lattice site can move to either of two otherwise equivalent neighboring sites. If both destinations offer the same opportunity for the next update, neither direction is favored. The modeling problem is to say, without relying on metaphor, what changes when one destination offers more opportunity for state change than the other.

The historical proposal answers in plain language: increase the probability of the next step toward the destination with greater opportunity. Established probability theory then requires a positive, normalized transition rule and asks what drift, diffusion, equilibrium, and continuum behavior follow. This paper calls the positive destination field **temporal opportunity** only after that operation is clear. QGTCD is the historical name for the larger proposal; the equations below test a bounded classical implementation of its directional atom rather than assuming the name already supplies quantum or gravitational physics.

Let $G = (V, E)$ be a locally finite undirected graph. A vertex x in V is a possible position and $N(x) = \{y : \{x, y\} \text{ in } E\}$ is its neighbor set. Let

$\tau : V \rightarrow (0, \infty)$

be a positive temporal-opportunity field. Only ratios of τ affect the kernel, so one may define a dimensionless field relative to a reference value,

$$u(x) = \log(\tau(x)/\tau_0).$$

Let β be a dimensionless response parameter and define

$$q(x) = (\tau(x)/\tau_0)^\beta = \exp(\beta u(x)).$$

The minimal endpoint-weighted transition law is

$$P_\beta(x, y) = \frac{1_{\{y \in N(x)\}} q(y)}{S(x)},$$

$$S(x) = \sum_{z \in N(x)} q(z).$$

This is a modeling choice made in 2026 to formalize the earlier mechanism. The historical prose specifies a directional probability bias but does not uniquely derive the endpoint weight, exponent, graph, or update time.

The sign of β fixes the direction of the tilt: $\beta > 0$ biases motion toward larger τ , $\beta = 0$ removes the field bias, and $\beta < 0$ biases motion toward smaller τ . The historical QGTCD direction therefore corresponds to $\beta > 0$.

Proposition 1 - positivity and normalization

If $\tau(x) > 0$ and x has finite nonzero degree, then $P_\beta(x, y) \geq 0$ and

$$\sum_y P_\beta(x, y) = 1.$$

Proof. Every endpoint weight $q(y)$ is positive. The neighbor sum $S(x)$ is therefore finite and positive. Dividing the nonnegative neighbor weights by their sum gives a normalized probability distribution. ■

Proposition 2 - constant-field recovery

If τ is constant on $N(x)$, then

$P_{\beta}(x,y) = 1/\deg(x)$ for every $y \in N(x)$.

On a regular lattice, the model therefore reduces exactly to the isotropic nearest-neighbor random walk when the temporal-opportunity field is locally constant. ■

This is the mathematical version of the first historical postulate: symmetric microscopic deviations cancel in a uniform environment.

2.1 The historical mechanism does not select a unique kernel

The 2022 source constrains the direction of the odds change: greater temporal opportunity should increase motion toward the corresponding neighboring state. It does not specify whether the update uses a destination softmax, a linear local response, an accept / reject rule, a self-loop, or a memory state. At least three plausible realizations satisfy the directional atom.

Kernel E: endpoint softmax

The endpoint realization uses

$$P_E(x,y) = 1_{\{y \in N(x)\}} \exp(\beta u(y)) / \sum_{z \in N(x)} \exp(\beta u(z)).$$

It is positive for all finite u , normalized, and reversible on finite undirected graphs with stationary mass proportional to $\exp(\beta u(x)) \sum_z \exp(\beta u(z))$. On a regular smooth lattice, its drift error is second order in the lattice spacing.

Kernel L: locally centered affine tilt

Let $k = \deg(x)$ and let $\bar{u}_N(x) = k^{-1} \sum_{z \in N(x)} u(z)$. Define

$$P_L(x,y) = 1_{\{y \in N(x)\}} k^{-1} [1 + \alpha (u(y) - \bar{u}_N(x))].$$

The centering makes the probabilities sum to one exactly. Positivity requires $1 + \alpha (u(y) - \bar{u}_N(x)) \geq 0$ for every neighbor, so this rule is local and cannot be used at arbitrarily large gradients without clipping or another completion. On a Cartesian lattice with a linear field, setting $\alpha = \beta$ gives the same leading drift as Kernel E exactly at every allowed mesh size. The price is a narrower admissible domain and no automatic global equilibrium law.

Kernel M: Metropolis accept / reject dynamics

On a finite periodic regular lattice, propose each nearest neighbor with probability $1/(2d)$ and target the stationary mass $\rho(x)$ proportional to $\exp(2\beta u(x))$. Accept a proposed move from x to y with

$$A(x,y) = \min\{1, \exp[2\beta (u(y) - u(x))]\},$$

and remain at x when the proposal is rejected. This is a standard Metropolis construction [16,17]. It differs from Kernel E by introducing self-loops and a nonsmooth accept / reject response. For a one-dimensional linear field it approaches the same leading drift with first-order, rather than second-order, finite-mesh error.

Property	Endpoint E	Centered L	Metropolis M
Globally positive for finite field	Yes	No; local positivity bound	Yes
Self-loop	No in the stated graph	No	Yes after rejection
Exact finite-graph reversible law	π proportional to q_S	Not automatic	ρ proportional to $\exp(2 \beta u)$ on the stated regular proposal graph
Linear-field drift error	$O(\epsilon^2)$	Zero while positivity holds	$O(\epsilon)$
Leading matched continuum drift	$2 \beta D \text{ grad } u$	$2 \beta D \text{ grad } u$	$2 \beta D \text{ grad } u$

Proposition 2A - matched leading local moments

On a d -dimensional Cartesian lattice with neighbors $x \pm e_i$, let u be four-times continuously differentiable on a compact neighborhood and set $\alpha = \beta$ in Kernel L. Where Kernel L satisfies its stated positivity bound, Kernels E and L have

$$E[\Delta X \mid X=x] = (\beta \epsilon^2/d) \text{ grad } u(x) + O(\epsilon^4)$$

and

$$E[\Delta X \Delta X^T \mid X=x] = (\epsilon^2/d) I + O(\epsilon^4).$$

Kernel M, with the stated symmetric proposal and target $\exp(2 \beta u)$, has

$$E[\Delta X \mid X=x] = (\beta \epsilon^2/d) \text{ grad } u(x) + O(\epsilon^3)$$

and

$$E[\Delta X \Delta X^T \mid X=x] = (\epsilon^2/d) I + O(\epsilon^3).$$

Under $\Delta t = \epsilon^2/(2dD)$, all three therefore have leading generator $D \Delta t + 2 \beta D \text{ grad } u \cdot \text{ grad}$; Kernel M retains an $O(\epsilon)$ finite-mesh remainder because of its accept / reject kink.

Derivation. For Kernel L, centering removes the constant and neighborhood- mean contributions to the first moment. Pairing opposite neighbors and Taylor expanding $u(x+\epsilon e_i) - u(x-\epsilon e_i)$ gives the stated first moment; pairing their second moments gives the stated covariance. For Kernel M, use $\min(1, \exp(r)) = 1 + \min(0, r) + O(r^2)$ with $r = 2 \beta [u(x+\delta) - u(x)]$. Opposite proposals cancel at zeroth order, their first-order acceptance imbalance supplies the same drift, and the rejection kink leaves the larger remainder. Dividing the moments by the diffusive time step yields the common leading generator. ■

The reproducible comparison in `model/qgtcd_draft2_kernel_comparison.py` checks normalization, positivity, finite-step nonidentity, convergence orders, and the common leading drift for a controlled one-dimensional linear field. It is a regression test of that case, not a numerical proof of the multidimensional statement above. The continuum drift is therefore not enough to identify the microscopic rule. Any physical QGTCD claim must either derive a kernel from deeper dynamics or measure an observable sensitive to finite-step structure.

3. Finite-graph equilibrium

The endpoint-weighted rule is a standard reversible weighted-graph or conductance-walk construction under a QGTCD interpretation. Define the symmetric edge conductance

$$c_{xy} = q(x)q(y)$$

for adjacent vertices. Then $P(x,y)=c_{xy}/\sum_z c_{xz}$. The mathematical novelty claimed here is not the invention of conductance walks; it is the explicit temporal-opportunity realization and the analysis of what that realization does and does not supply [13].

Proposition 3 - detailed balance and stationary measure

Assume G is finite, connected, undirected, and has at least two vertices (and therefore at least one edge). Define

$$\pi(x) = \frac{q(x)S(x)}{\sum_v q(v)S(v)},$$

Then π is a probability distribution and satisfies detailed balance:

$$\pi(x)P_{\beta}(x,y) = \pi(y)P_{\beta}(y,x)$$

for every edge $\{x, y\}$. Hence π is stationary.

Proof. Normalization follows from positivity and the definition of Z . For adjacent vertices,

$$\begin{aligned} \pi(x)P_{\beta}(x,y) &= \frac{q(x)S(x)}{Z} \frac{q(y)}{S(x)} \\ &= \frac{q(x)q(y)}{Z} \\ &= \pi(y)P_{\beta}(y,x). \end{aligned}$$

Summing detailed balance over incoming neighbors establishes stationarity. ■

On a uniform lattice with cell volume ϵ^d , $\pi(x)$ is a probability mass at a grid point, not yet a continuum density. For a smooth field away from boundaries, $S_{\epsilon}(x) = 2d q(x) + \epsilon^2 \Delta q(x) + O(\epsilon^4)$, so the corresponding piecewise-constant cell density approaches a normalized density proportional to $q(x)^2 = \tau(x)^{2\beta}$. Boundaries, nonuniform cell volumes, and rapid spatial variation modify that approximation; the exact graph formula remains π proportional to qS .

This equilibrium is a property of the chosen Markov process. It is not yet an energy law. No thermodynamic interpretation is asserted without an explicit energy, bath, and entropy-production model.

4. Regular-lattice drift and generator asymptotic

Consider a d -dimensional Cartesian lattice with spacing ϵ and neighbors $x \pm \epsilon e_i$. Assume u is sufficiently smooth. Taylor expansion of $q = \exp(\beta u)$ gives

$$E[\Delta X \mid X=x] = (\beta \epsilon^2/d) \text{grad } u(x) + O(\epsilon^4)$$

and

$$E[\Delta X \Delta X^T \mid X=x] = (\epsilon^2/d) I + O(\epsilon^4).$$

Choose the diffusive time scaling

$$\Delta t = \epsilon^2/(2dD),$$

where $D > 0$ has dimensions of diffusion, $\text{length}^2/\text{time}$.

Proposition 4A - discrete-generator asymptotic

Let u, f be four-times continuously differentiable on $\mathbb{R}^d$, with derivatives through fourth order bounded on compact neighborhoods. Let $D > 0$ and finite β in $\mathbb{R}$. Define P_ϵ on $\mathbb{Z}^d$ by the endpoint-weight rule and $\Delta t = \epsilon^2 / (2dD)$. Then, locally uniformly,

$$\begin{aligned} L_\epsilon f(x) &= E_x[f(X_{(n+1)}) - f(X_n)] / \Delta t \\ &= D \text{Laplacian}(f)(x) + 2\beta D \text{grad } u(x) \cdot \text{grad } f(x) + O(\epsilon^2). \end{aligned}$$

The differential operator on the right is the generator of the candidate diffusion, whenever the drift satisfies conditions ensuring a well-posed stochastic differential equation,

$$dX_t = \sqrt{2\beta D} \text{grad} \log(\tau(X_t) / \tau_0) dt + \sqrt{2D} dW_t,$$

with candidate density equation

$$\begin{aligned} \partial_t p &= D \text{Laplacian}(p) \\ &- \text{div}[2\beta D p \text{grad} \log(\tau / \tau_0)]. \end{aligned}$$

Weak convergence of interpolated chains does not follow from the local expansion alone; it requires a separate diffusion-approximation theorem. Section 4.1 supplies that argument for a compatible periodic compact domain [14].

Derivation. Insert the first two conditional moments into the Taylor expansion of $E[f(X + \Delta X) - f(X)] / \Delta t$. The first moment supplies the drift term and the second supplies $D \text{Laplacian}(f)$; the remaining generator terms are $O(\epsilon^2)$ under the stated interior smoothness assumptions. This proves the generator expansion but not, by itself, convergence of whole sample paths. ■

Thus the minimal kernel converts a gradient in log temporal opportunity into a drift coefficient or conditional mean displacement rate

$$b(x) = \sqrt{2\beta D} \text{grad} \log(\tau(x) / \tau_0).$$

This result captures the historical directional atom with a precise caveat: the field entering the drift is $\text{grad} \log(\tau / \tau_0)$, not automatically $\text{grad } \tau$, and the proportionality contains the microscopic diffusion scale D .

4.1 Process convergence on a periodic compact domain

The local generator expansion was proved first but correctly left weak path convergence open. That gap can be closed for one clean boundary-free setting.

Proposition 4B - weak convergence of the endpoint chain on a torus

Let the state space be the unit d -dimensional flat torus T^d , discretized by compatible periodic Cartesian lattices with $\epsilon_n = 1/N_n \rightarrow 0$ for integers N_n . Let u in $C^4(T^d)$, $D > 0$, and finite β . Run Kernel E with time step $\Delta t_n = \epsilon_n^2 / (2dD)$, and define the piecewise-constant interpolation by $X^n(t) = X^n_{\text{floor}(t/\Delta t_n)}$. If the initial laws converge weakly, then for every finite T ,

$$X^n \Rightarrow X \text{ in } D([0, T], T^d),$$

where X is the unique diffusion solving

$$dX_t = \sqrt{2\beta D} \text{grad} u(X_t) dt + \sqrt{2D} dW_t$$

on the torus, equivalently the martingale problem with generator

$$L f = D \Delta f + 2 \beta D \operatorname{grad} u \cdot \operatorname{grad} f.$$

Proof. Proposition 4A gives uniform convergence $L_n f \rightarrow Lf$ for every periodic f in $C^4(T^d)$ because the domain is compact and all required derivatives are bounded. Every jump has size $\epsilonpsilon_n$, so the maximum jump vanishes. Compact containment is automatic on T^d . The limiting drift is smooth and bounded and the diffusion matrix is constant and uniformly elliptic; consequently the martingale problem for L is well posed. The generator-convergence theorem for Markov processes then gives weak convergence of the interpolated chains [14]. ■

This theorem is intentionally narrow. It does not cover a point-source singularity, noncompact space, absorbing or reflecting walls, irregular graphs, a dynamic metric, or the persistent-velocity model proposed later. Each change requires its own domain, tightness, and boundary analysis.

Proposition 4C - invariant-density and stationary-measure consistency

Under the assumptions of Proposition 4B, the limiting diffusion has the unique invariant density

$$p_{\text{infinity}}(x) = \exp(2 \beta u(x)) / \int_{T^d} \exp(2 \beta u(z)) dz.$$

It is reversible with respect to this density. If π_n is the exact stationary mass of Kernel E on the $\epsilonpsilon_n$ grid and ρ_n is the piecewise-constant density that equals $\pi_n(x)/\epsilonpsilon_n^d$ on the cell associated with x , then ρ_n converges in $L^1(T^d)$ to p_{infinity} .

Proof. For the limiting diffusion, the probability current at a density p is $J = 2\beta D \nabla p \cdot \operatorname{grad} u - D \operatorname{grad} p$. Substitution of p_{infinity} gives $J = 0$, so the density is stationary and satisfies detailed balance. Smooth coefficients and uniform ellipticity on the connected compact torus give uniqueness. For the chain, $\pi_n(x) \propto q(x) S_n(x)$, while the symmetric-neighbor expansion gives $S_n(x) = 2d q(x) + \epsilonpsilon_n^2 \Delta q(x) + O(\epsilonpsilon_n^4)$ uniformly. Hence $q(x) S_n(x) = 2d \exp(2 \beta u(x)) [1 + O(\epsilonpsilon_n^2)]$. Normalization is a Riemann sum, and smoothness on the compact torus makes the induced cell densities converge in L^1 to the stated limit. QED.

This agreement is a mathematical consistency relation between the selected finite chain and its diffusion limit. It does not provide an energy, temperature, heat bath, entropy-production law, or physical evidence that the stationary density describes gravitational matter.

4.2 Exact one-dimensional check

For $\tau(x) = \tau_0 \exp(\kappa x)$ in one dimension, where κ has units $1/\text{length}$,

$$P(+\epsilonpsilon) - P(-\epsilonpsilon) = \tanh(\beta \kappa \epsilonpsilon).$$

Therefore the exact rescaled drift is

$$b_{\epsilonpsilon} = [\epsilonpsilon / \Delta t] \tanh(\beta \kappa \epsilonpsilon),$$

which approaches $2 \beta D \kappa$ under the scaling above. The accompanying reproducibility script checks this exact expression, normalization, regular-lattice isotropy, the finite-path stationary measure, and detailed balance.

5. Conditional Newtonian comparison

Let $\Phi(x)$ be the Newtonian gravitational potential per unit mass, with units $\text{length}^2/\text{time}^2$, and let $\tau_r > 0$ be a response or relaxation time. In the overdamped comparison law $b = -\tau_r \operatorname{grad} \Phi$, the coefficient τ_r converts acceleration units into drift-velocity units. If one chooses the field map

$$u(x) = \log(\tau/\tau_0) = -t_r \Phi(x)/(2\beta D),$$

then the stochastic drift becomes

$$b(x) = -t_r \nabla \Phi(x).$$

For a point source outside the source region,

$$\Phi(r) = -GM/r,$$

so

$$u(r) = C + t_r GM/(2\beta D r)$$

and

$$b(r) = -t_r GM/r^2 \hat{r}.$$

Proposition 5 - conditional inverse-square embedding

Under the specified field map, the continuum drift has the inward inverse-square radial form outside a Newtonian point source. ■

This proposition is exact as an algebraic substitution and limited as a physical result: inverse-square behavior follows because the Newtonian potential was placed into u by choice. It does not derive τ from mass, explain why the mapping is universal, or turn first-order overdamped motion into the second-order inertial equation

$$d^2X/dt^2 = -\nabla \Phi.$$

A viable QGTCD gravity model therefore needs an inertial extension - such as a velocity-state or persistent-jump process - or a principled account of why an overdamped effective description applies. Simply relabeling b as acceleration would be dimensionally wrong. If u is a universal source field, the combination $t_r/(\beta D)$ cannot be allowed to vary arbitrarily between probes while describing the same field.

6. What the minimal model reveals

6.1 A free-fall universality obligation

The drift coefficient is $2\beta D$. If different particles or preparations have different microscopic diffusion constants, the unmodified kernel predicts different drifts in the same τ field unless β , the field map, or a more fundamental dynamics compensates exactly. Universal free fall is therefore not an automatic success of the model; it is a sharp construction and falsification gate.

6.2 Equilibrium does not equal gravity

The finite graph accumulates probability according to π proportional to qS . That can model biased occupancy. Gravitational systems, however, involve inertial trajectories, conserved quantities, many-body sources, and dynamical geometry. An equilibrium distribution alone is not a gravitational field equation.

6.3 Probability is not yet quantum probability

The kernel evolves nonnegative classical probabilities. It has no complex amplitude, unitary evolution, interference phase, Hilbert-space observable, or Born-rule derivation. A quantum extension would need to state whether the kernel is an emergent open-system limit, a rule for amplitudes, or a stochastic modification of quantum dynamics, and then show normalization and empirical compatibility in that chosen framework.

6.4 The field needs an operational definition

τ is presently a positive model field. It could be related to a local clock rate, proper-time increment, state-counting measure, or another observable, but those possibilities are not interchangeable. The next physical draft must define how τ is measured, transformed between frames, sourced by matter, and evolved in time.

6.5 Which results depend on which assumptions

Result	Regular lattice	Finite graph	Boundary rule	Smooth field	Compactness
Positivity and normalization	No	No	Only a nonempty finite neighbor set	No	No
Constant-field simple walk	No	No	Degree determines the local walk	No	No
Detailed balance and normalized stationary mass	No	Yes, connected and undirected	Encoded by the graph	No	Finiteness supplies normalization
Generator asymptotic	Yes, Cartesian in the kernel comparison	No	Interior or periodic statement	C^4 locally	No for the local statement
Weak path convergence in Proposition 4B	Yes, periodic Cartesian	No	Periodic torus only	C^4	Yes
Conditional inverse-square embedding	Euclidean exterior comparison	No	Source region excluded	Smooth away from source	No

Reflecting boundaries would change the neighbor rule and normally lead to a Neumann-type limiting condition; absorbing boundaries would kill paths; irregular lattices can produce anisotropic or position-dependent diffusion. None is a harmless implementation detail if QGTCD is proposed as a physical theory.

6.6 Covariance obligations

The current generator uses Euclidean space, a preferred update time, and a scalar field u . It is coordinate-covariant only under ordinary Euclidean changes of coordinates. A natural fixed-spatial- geometry analogue would be

$$L_g f = D \Delta_g f + 2 \beta D \langle \text{grad}_g u, \text{grad}_g f \rangle_g,$$

where Δ_g is the Laplace-Beltrami operator. This expression is invariant under spatial coordinate changes on a prescribed Riemannian manifold. It does **not** supply spacetime diffeomorphism invariance, determine g , remove the preferred Markov time, or couple consistently to energy and momentum. A covariant physical version must specify the spacetime state, evolution parameter, source equation, matter action or stochastic law, and conservation identity.

Established equivalence-principle and post-Newtonian tests define the empirical constraint family summarized below [12].

6.7 Equivalence-principle and probe-independence obligations

Obligation	Operational question	Current status
Weak equivalence	Do compositionally different freely falling probes follow the same trajectory?	Open; b contains D and β
Local Lorentz invariance	Can a local nongravitational experiment detect the preferred update frame?	Open; the model has an explicit Markov time
Local position invariance	Do nongravitational outcomes depend only on local standards rather than absolute location?	Open; τ lacks an operational clock protocol
Strong equivalence	Does gravitational self-energy couple universally?	Not modeled

Obligation	Operational question	Current status
Probe independence	Do clocks, massive matter, light, and composite systems infer the same τ ?	Not modeled
Source independence	Is the field determined by source state rather than by the test probe's D , β , or preparation?	Not modeled

Calling τ “time dilation” cannot satisfy these obligations. The theory needs a measurement procedure for τ , separate response laws for clocks, massive matter, and light, and a demonstration that any probe parameters cancel from universal free fall.

6.8 A discriminating observable: fixed-time response-diffusivity slope

The minimal endpoint diffusion predicts, at fixed u and β ,

$$|b| = 2 |\beta| D |\text{grad } u|.$$

Drift and acceleration have different dimensions, so they cannot be entered as though they were the same measured response. Define instead one common observable. For a preregistered interval T , subtract the matched zero-gradient baseline and measure the mean field-aligned displacement $R_D(T)$. In a region where the field gradient is approximately constant, the overdamped endpoint model gives

$$R_{OD}(T) = 2\beta D (\partial_{\parallel} u) T,$$

whereas the inertial calibration, with matched initial field-aligned velocity, gives

$$R_{IN}(T) = (1/2) c_*^2 (\partial_{\parallel} u) T^2.$$

Define the preparation-dependence slope using that same displacement observable,

$$s_D = \partial \log |R_D(T)| / \partial \log D.$$

The unrenormalized minimal model predicts $s_D=1$. A universal-field free-fall description predicts that gravitational response does not vary with an arbitrarily tunable nongravitational diffusion coefficient, corresponding to $s_D=0$ after Brownian, thermal, drag, and apparatus effects are removed. This gives a genuine discriminator rather than a reinterpretation of an already fitted inverse-square curve.

A controlled test would hold the independently measured temporal-field proxy fixed, vary the effective diffusivity of otherwise matched neutral probes, reverse the field geometry, and fit $\log |R_D(T)| = c + s_D \log D$ with blinded analysis. Ordinary diffusion and drag can easily create an apparent slope, so a credible signal must survive zero-gradient controls, thermal and electrostatic controls, orientation reversal, and independent clock mapping. A result near zero would falsify the unrenormalized endpoint kernel as a universal free-fall mechanism. A result near one would still not establish gravity: it would first need exclusion of all ordinary stochastic forces and replication with compositionally different probes.

The kernel-comparison script verifies only the algebraic model prediction that the endpoint drift ratio, and therefore fixed-time mean displacement at common T , equals the imposed diffusivity ratio. It provides no empirical result.

7. Physical obligation and falsifier matrix

Requirement	Minimal-kernel result	What remains	Failure condition
Positive normalized transitions	Proved for positive τ and finite degree	Derive or constrain the kernel physically	Negative weights, divergent neighbor sum, or empirical transition rule incompatible with the tilt
Uniform-field isotropy	Exact on a regular symmetric lattice	Define the appropriate local inertial frame	Preferred direction in a truly uniform field after controls
Directional drift	$b = 2\beta D \text{grad} \log(\tau/\tau_0)$	Measure or derive τ and coupling	Wrong sign or gradient dependence
Inverse-square behavior	Algebraically embedded through a chosen τ - Φ map	Derive the map from source dynamics	No admissible universal map compatible with observations
Newtonian free fall	Not supplied; model is overdamped	Add inertial dynamics and universal coupling	Unavoidable species-, state-, or temperature-dependent free fall above bounds
Equivalence principle	Open	Demonstrate local universality and redshift consistency	Composition or internal-state dependence inconsistent with experiment
Relativistic covariance	Open	Define spacetime field and transformation law	Preferred-frame effects above bounds or acausal propagation
Light bending and Shapiro delay	Open	Specify null propagation	Cannot reproduce lensing and time-delay constraints simultaneously
Conservation and backreaction	Open	Supply source and field dynamics	Probability, energy, or momentum accounting fails
Strong-field / gravitational waves	Open	Nonlinear covariant completion	Conflict with binary-pulsar, black-hole, or wave observations
Distinguishing prediction	Not yet frozen	Derive a quantitative residual beyond GR	Model has no non-fitted observable difference or is excluded by null tests

8. Candidate next model: persistent velocity states

The smallest route from drift to acceleration is to enlarge the state from position X_n to position and velocity (X_n, V_n) . Temporal measure could bias changes in velocity while preserving an inertial continuation probability. A candidate family would have to specify

$$P(V_{n+1}=v' \mid X_n=x, V_n=v)$$

with four non-negotiable limits:

1. inertial persistence when $\text{grad } \tau = 0$;
2. an acceleration proportional to $\text{grad} \log(\tau/\tau_0)$ rather than a terminal drift;
3. a particle-independent weak-field limit;
4. a controlled continuum equation with conserved probability.

This extension is not supplied here because choosing it without a source or derivation would replace one underdetermined kernel with another. It is the next mathematical problem exposed by the minimal model.

9. Legacy-equation firewall

Later QGTCD notes explored energy corrections, modified Schrödinger equations, metric modifications, cosmology, and interactions with Super Dark Time and Super Information Theory. They are part of the

theory's genealogy but are not load-bearing in this paper. Several require correction before reuse:

- $m = h/(Nf)$ is dimensionally inconsistent with mass and does not follow from $mc^2=hf/N$; that neighboring equation would instead give $m=hf/(Nc^2)$.
- For positive k , $mc^2 + k/\rho_t$ approaches mc^2 from above, not below.
- For $\rho_t>0$ and positive α and k , $mc^2 - \alpha \rho_t + k/\rho_t$ is strictly decreasing because its derivative is $-\alpha - k/\rho_t^2$; it does not produce the claimed dip-and-rise form.
- Rearranging a modified Schrödinger equation while the unknown wavefunction and its time derivative remain on the right is not a solved wavefunction.

These observations preserve the historical equation record while preventing an inconsistent later branch from weakening the independent 2022 stochastic atom. A separate appendix will date and type every legacy equation as historical, dimensionally unresolved, contradicted, repaired, or superseded.

10. Relation to Super Dark Time and Super Information Theory

QGTCD, Super Dark Time (SDT), and Super Information Theory (SIT) have an intellectual genealogy, but they are not interchangeable evidence. The transition kernel formalized here is a QGTCD-specific mechanism. Later SDT and SIT field expressions do not independently validate it, and a result in the formal SIT proof program cannot be imported as proof of this physical model. Conversely, failure of a later energy formula would not erase the earlier probability-distortion mechanism; it would identify a failed descendant branch.

The correct relationship is therefore versioned and typed:

- QGTCD supplies the 2022 stochastic temporal-measure proposal;
- SDT develops a later time-density field direction;
- SIT develops later informational and field-theoretic directions;
- each model keeps its own equations, assumptions, tests, and proof status.

11. Conditional inertial calibration

The endpoint kernel cannot be identified directly with universal free fall without an exact compensation for the probe diffusion scale. The calibration therefore separates two questions that the historical prose joined: how temporal opportunity might bias microscopic transitions, and how a universal external field would accelerate a freely falling body. The resulting construction is a classical position-velocity benchmark. Because its field map, source law, and response law are imposed, it is not derived from the endpoint kernel and is not a new account of gravity.

11.1 Operational semantics of the field

Let

$$u(x,t) = \log[\tau_{op}(x,t)/\tau_0]$$

be a dimensionless **dilation-depth** or temporal-opportunity field. The notation τ_{op} must not be confused with the proper time read by a local clock. Near a positive mass, the historical mechanism requires greater temporal opportunity in the inward direction. In the Newtonian exterior benchmark this means $u=GM/(c_*^2 r)>0$, which increases toward the source. A stationary clock deeper in the same potential runs more slowly relative to a distant coordinate clock. Thus the two quantities vary oppositely at leading weak-field order.

This distinction repairs an ambiguity in earlier QGTCD language. “More time” can mean a greater proposed density of microscopic temporal opportunities while an ordinary clock accumulates less proper time per unit distant coordinate time. A physical theory must define which quantity an instrument measures rather than moving between the two meanings.

11.2 External-field source benchmark

Let Φ be the Newtonian potential per unit mass, with

$$\Delta\Phi = 4\pi G\rho_m,$$

and define

$$u = -\Phi/c_*^2. \quad (1)$$

Here c_* is a universal conversion scale with dimensions of speed. The field then obeys

$$\Delta u = -4\pi G\rho_m/c_*^2. \quad (2)$$

For a point source, $\Phi = -GM/r$ and $u = GM/(c_*^2 r)$. Equations (1)-(2) do not derive a new source law. They rewrite Newtonian gravity in terms of a dimensionless positive dilation-depth field. Their purpose is to expose the exact additional assumptions needed before QGTCD language yields a physical acceleration.

11.3 Position-velocity dynamics

Enlarge the state from position to phase space (X_t, V_t) and define

$$dX_t = V_t dt,$$

$$dV_t = c_*^2 \text{grad } u(X_t, t) dt + \sigma_v dW_t. \quad (3)$$

The parameter σ_v has units of acceleration times the square root of time and represents an optional random-acceleration sector. It is not required by the universal field response and should be set to zero in the minimal free-fall benchmark. Crucially, the deterministic acceleration coefficient is universal and contains neither the position-diffusion coefficient D nor a probe-specific response time.

Proposition 6 - Conditional Newtonian equivalence identity

Assume $\sigma_v = 0$, a static field, and the mapping (1). Then

$$d^2 X_t / dt^2 = c_*^2 \text{grad } u(X_t) = -\text{grad } \Phi(X_t). \quad (4)$$

All probes assigned the same initial position and velocity follow the same trajectory independently of mass, composition, and the overdamped diffusion parameter. In the point-source exterior this gives

$$d^2 X_t / dt^2 = -GM \hat{r} / r^2.$$

Proof. Substitute $u = -\Phi/c_*^2$ into (3) and differentiate the position equation. The factors c_*^2 cancel. The resulting acceleration is independent of probe properties. The point-source expression follows from $\Phi = -GM/r$. QED.

The proposition passes the Newtonian weak-equivalence benchmark only because a universal inertial law and the Newtonian source map have been imposed. Universality is therefore a property of the postulate, not a derived consequence of the 2022 probability narrative.

Non-derivation lemma. Equations (1) and (3) jointly encode the Newtonian result: the first defines u from the already-known potential and the second selects the cancelling response coefficient. Their substitution

cannot independently explain why mass sources Φ , why the response is universal, or why the coefficient is c^2 . The identity is valuable as a consistency and bookkeeping result, but it supplies no new force law.

For a static field and $\sigma_v=0$, the specific mechanical energy

$$E_{\text{spec}} = |V|^2/2 - c^2 u = |V|^2/2 + \Phi \quad (5)$$

is conserved along solutions. This follows from $dE_{\text{spec}}/dt = V \cdot c^2 \text{grad } u - c^2 \text{grad } u \cdot V = 0$. With $\sigma_v>0$, the noise injects velocity variance and the same pathwise conservation law no longer holds; an energy budget or bath model would then be required.

The phase-space density $f(x, v, t)$ obeys the Kramers-type equation

$$\begin{aligned} \text{partial}_t f + v \cdot \text{grad}_x f + c^2 \text{grad } u \cdot \text{grad}_v f \\ = (\sigma_v^2/2) \Delta_v f. \quad (6) \end{aligned}$$

Under sufficient decay in velocity and no-flux or periodic spatial conditions, integrating (6) over phase space conserves total probability. This supplies a consistent probability evolution, but not a quantum amplitude, stress-energy tensor, gravitational conservation law, or equilibrium distribution.

Additive-noise energy audit

In the force-free case, each velocity component satisfies $dV_i = \sigma_v \, dW_i$, so $\text{Var}[V_i(t)] = \text{Var}[V_i(0)] + \sigma_v^2 t$. The expected kinetic energy therefore grows linearly rather than relaxing to a stationary value. This stochastic sector is mathematically normalized but physically incomplete. A thermodynamic interpretation would require friction, a specified bath, a fluctuation-dissipation relation, or an explicit external energy source. No dissipative term is added merely to make the process look complete.

11.4 Weak-field clock relation

In general relativity, a stationary weak-field clock satisfies

$$d\tau_{\text{proper}}/dt \text{ approximately } 1 + \Phi/c^2.$$

If the universal conversion scale is identified with light speed, $c=c$, then (1) gives

$$d\tau_{\text{proper}}/dt \text{ approximately } 1 - u. \quad (7)$$

Equivalently, after imposing the GR weak-field clock map, gravitational redshift estimates the benchmark variable $-u$ through the local clock rate. This creates a consistency test: the u inferred from clock comparisons and the u inferred from free-fall acceleration must agree without probe-dependent fitting. It does **not** independently measure the proposed microscopic quantity τ_{op} , because in this calibration both the clock relation and $u = -\Phi/c^2$ were imported from the same standard weak-field structure. An independent QGTCD operationalization must identify an observable not defined by first reconstructing the known potential.

Equation (7) is only the leading weak-field relation. QGTCD does not yet derive the higher-order clock law, moving-clock terms, rotating-source terms, or a coordinate-independent proper-time functional.

11.5 Light propagation and the PPN gap

Matching Newtonian acceleration and first-order clock redshift does not determine light bending or Shapiro delay. A common weak-field metric parameterization can be written schematically as

$$ds^2 = -(1-2u)c^2 dt^2 + (1+2 \gamma_{\text{PPN}} u) \delta_{ij} dx^i dx^j + O(u^2). \quad (8)$$

The clock relation constrains the leading temporal component. Light deflection and time delay also depend on spatial curvature through γ_{PPN} . General relativity gives $\gamma_{\text{PPN}}=1$, while a model that changes only the time component generally does not recover the full result. The leading deflection for impact parameter b is proportional to

$$2(1+\gamma_{\text{PPN}})GM/(b c^2). \quad (9)$$

The calibration does not derive (8), select γ_{PPN} , or establish that photons follow null geodesics of any QGTCD metric. Consequently it cannot yet claim lensing, Shapiro delay, perihelion advance, frame dragging, or gravitational-wave agreement.

This is the central relativistic bridge: a temporal field alone is not a spacetime theory. QGTCD must either derive a metric and matter action, define a nonmetric but covariant propagation law, or remain an explicitly Newtonian effective model.

11.6 Source dynamics, backreaction, and conservation

The Poisson benchmark is instantaneous and static. A physical completion must answer:

1. Does u propagate, and at what speed?
2. What source quantity replaces Newtonian mass density in relativistic settings?
3. How do pressure, stress, field energy, and gravitational self-energy contribute?
4. What conservation identity links matter exchange to field evolution?
5. Does the theory introduce a preferred frame through its update time?

Promoting (2) mechanically to a scalar wave equation would create a scalar-gravity candidate, not general relativity. Such models face strong light-propagation, binary-pulsar, equivalence, and post-Newtonian constraints. The model therefore leaves the dynamic source equation open instead of selecting one for cosmetic completeness.

11.7 Internal-model comparison

The overdamped endpoint diffusion predicts $s_D=1$ for the fixed-time, baseline-subtracted mean displacement defined in Section 6.8. The inertial bridge (3), evaluated over the same interval with matched initial velocity, predicts a deterministic mean displacement independent of D , hence $s_D=0$, while allowing an optional stochastic spread through σ_v . This is a comparison of the same observable under two materially different models, not a comparison of drift units with acceleration units and not a parameter adjustment.

A preparation-dependence experiment can therefore discriminate two internal QGTCD implementations:

- fixed-time mean displacement varying with controlled diffusivity as predicted by the overdamped implementation supports neither universal free fall nor the inertial bridge and would require extraordinary control of ordinary forces;
- no such dependence rejects the unrenormalized overdamped kernel as the direct gravity law but is compatible with the inertial bridge and with standard gravity;
- agreement with the inertial bridge is not evidence for QGTCD unless an additional non-GR prediction also succeeds.

The last point is decisive. By imposing $u=-\Phi/c_*^2$, the inertial calibration reproduces standard Newtonian behavior. The model earns physical novelty only if the temporal-opportunity interpretation yields a quantitative residual not already fixed by general relativity and survives existing bounds.

11.8 Quantum status

Nothing in (1)-(9) is quantum. W_t is classical noise, and setting $\sigma_v=0$ leaves a deterministic classical trajectory. The historical name “Quantum Gradient Time Crystal Dilation” does not alter that classification. A future quantum descendant would need a Hilbert-space state, an evolution law, observables, a measurement rule, a time-crystal order parameter, and a controlled classical limit. It would also need to explain whether the stochastic transition mechanism is fundamental, emergent, or an open-system approximation.

11.9 Bridge audit

Physical requirement	Present result	Remaining status
Inertial continuation	Explicit position-velocity state	supplied for the benchmark
Universal weak-field acceleration	$a=c_*^2 \text{grad } u = -\text{grad } \Phi$	supplied by imposed universal mapping
Independence from probe diffusivity	deterministic acceleration contains no D	supplied for the benchmark
Static energy accounting	$E_{\text{spec}} = \text{norm}(V)^2/2 - c_*^2 u$ conserved when $\sigma_v=0$	supplied conditionally
Probability conservation	Kramers equation under stated boundaries	supplied conditionally
Operational clock relation	$d\tau_{\text{proper}}/dt \approx 1 - u$ if $c_*=c$	weak-field compatibility, not derived
Matter source	Poisson benchmark	imported from Newtonian gravity
Spatial curvature and light	PPN parameter remains free	open
Dynamic covariance	no spacetime action or field equation	open
Backreaction and stress-energy conservation	not modeled	open
Distinctive prediction beyond GR	none in this bounded endpoint	required only for a future physical theory
Quantum dynamics	absent	historical label only

The coherent inertial benchmark has a deliberately bounded epistemic status: it is a conditional equivalence and calibration construction. Once universality, clocks, and Newtonian sourcing are imposed, compatibility with standard weak-field physics is expected. The construction has not produced a new gravitational theory.

11.10 Publication bifurcation

The present work supports two different research products that should not be conflated.

Track A: bounded classical model paper. This manuscript is the completed local endpoint of the present preprint: a source-traced study of temporal-opportunity kernels, their continuum equivalence class, the failure of a direct overdamped gravity interpretation, and a conditional inertial calibration. Independent mathematical and physics review remains valuable but does not require adding unsupported physics to this bounded paper.

Track B: future gravity theory. This route requires a dynamical principle or action, a derived source coupling, matter and light propagation, conservation and backreaction, covariance or a constrained preferred-frame structure, an actual quantum sector if the label is scientific, and a preregistered quantitative prediction that differs from general relativity.

Track B is not reachable by interpretive expansion of the present equations. It is not part of this preprint’s release gate. Until its separate obligations are met, physical claims in this paper remain inside Track A. # 12. Empirical discrimination program

The paper has one presently testable comparison, but it is an internal model-selection test rather than a test of QGTCD against general relativity. Using the same fixed-time, baseline-subtracted mean displacement,

the overdamped implementation predicts $s_D=1$, whereas the inertial calibration predicts $s_D=0$ because its deterministic response coefficient is imposed independently of D .

12.1 Measured quantities and estimand

An experiment must measure the probe's effective diffusion coefficient D independently of the field-directed response. Choose a common preregistered interval T , match or estimate the initial field-aligned velocity, subtract the corresponding zero-gradient or sham-source baseline, and estimate the signed mean displacement $R_j(T)$ along the prespecified field axis. Fit

$$\log |R_j(T)| = \alpha + s_D \log D_j + \epsilon_{j,T}.$$

The primary estimand is s_D , with uncertainty propagated from both D_j and $R_j(T)$. The chosen interval must be short enough for the local-gradient approximation or else the full spatial model must be fitted. The temporal-field proxy must be measured independently or manipulated through a preregistered source geometry; it may not be reconstructed from the response being tested.

12.2 Minimum control structure

1. Use at least three independently characterized diffusivity preparations spanning a prespecified range.
2. Include zero-gradient, source-reversal, apparatus-reversal, and sham-source conditions.
3. Monitor thermal, electrostatic, magnetic, pressure, vibration, and drag covariates.
4. Use the same observation interval and displacement estimator across preparations.
5. Blind the response fit to preparation labels and reserve the model-selection rule before unblinding.
6. Replicate any surviving effect with compositionally different neutral probes and an independently built apparatus.

12.3 Decision rule and failure conditions

An estimate statistically and practically compatible with $s_D=0$ rejects the unrenormalized overdamped kernel as the direct universal free-fall law; it does not support QGTCD because standard gravity and the imported inertial calibration also predict no arbitrary diffusivity dependence. An estimate compatible with $s_D=1$ remains uninterpretable as gravity unless ordinary stochastic forces are excluded and the effect tracks source reversal. A slope that changes across apparatus, temperature, or composition fails the claimed universal mechanism. Any field proxy defined from the known Newtonian potential rather than independently measured fails the operational test before data analysis.

A future physical-QGTCD test must add a quantitative residual that differs from both general relativity and ordinary environmental-force models. No such residual is claimed here.

13. Limitations and failure boundaries

13.1 Mathematical scope

The continuum result uses smooth fields and matched lattice scaling. The path-convergence result is restricted to a periodic compact domain. Boundary conditions, irregular graphs, anisotropic neighborhoods, singular source regions, and state-dependent update times can change the limit. The model family is therefore neither unique nor geometry-free.

13.2 Physical scope

The temporal-opportunity field has no independent operational measurement protocol. The Newtonian source equation, coupling constant, and universal response are imported in the calibration. No action, stress-energy source, backreaction law, conservation identity, or controlled preferred-frame test is supplied. The paper cannot adjudicate lensing, Shapiro delay, perihelion advance, frame dragging, binary dynamics, or gravitational radiation.

13.3 Stochastic and thermodynamic scope

The position-only kernels define normalized stochastic dynamics and, on the compact periodic domain, a reversible invariant probability density. That mathematical stationarity is not a thermodynamic equilibrium claim. No physical energy, bath, temperature, or entropy-production model has been specified. In the inertial model, additive velocity noise without friction heats the velocity distribution. A physical bath model would need specified dissipation and fluctuation-dissipation structure; adding those terms would define a new model requiring a new audit.

13.4 Quantum and time-crystal scope

No Hilbert space, quantum state, operator algebra, measurement rule, objective time-crystal order parameter, or controlled quantum-to-classical limit appears in the formalism. The words “quantum” and “time crystal” remain part of the historical theory name and genealogy, not a scientific classification of the equations proved here.

13.5 Novelty boundary

The source genealogy supports the documented development of the temporal-opportunity probability mechanism. It does not establish physical correctness. The conditional Newtonian identity is a transparent calibration, not a novel gravitational prediction. Scientific novelty would require a separately derived and empirically viable consequence not already fixed by the imported Newtonian or relativistic structure.

14. Discussion

The formalization converts a historical narrative into explicit models that can be compared, derived, and rejected. A positive temporal-opportunity field can tilt local transitions; a uniform field recovers an unbiased walk; and several finite-step rules share a leading continuum drift. For the endpoint rule on compatible periodic lattices, finite stationary measures also converge to the limiting diffusion’s reversible invariant density. These are substantive mathematical clarifications because they identify a consistent model class and an identifiability problem: a measured leading drift or stationary profile would not by itself select the microscopic kernel.

The same analysis prevents an attractive analogy from doing the work of a physical theory. The position-only process is overdamped and preparation-dependent. The position-velocity calibration recovers Newtonian acceleration only after the Newtonian potential and a cancelling universal response coefficient are supplied. Leading weak-field clock compatibility is likewise imported. These are useful consistency checks because they reveal the missing source, matter, light, covariance, and conservation structures, but they are not evidence that temporal opportunity causes gravity.

The bounded model-paper route and a future gravity-theory route must remain separate. The present paper can be reviewed on the accuracy of its genealogy, stochastic definitions, asymptotics, conditional identities,

and proposed internal discriminator. A future gravity theory would require materially new mathematics and physics, not a more assertive interpretation of the same equations.

15. Conclusion

QGTCD's documented core proposes that unequal temporal opportunity changes directional odds. This paper gives that proposal a conservative mathematical form. Three distinct finite-step kernels share a leading drift-diffusion limit under matched assumptions, while retaining observably different finite-step behavior. The endpoint kernel's stationary measures converge to the reversible invariant density of its limiting diffusion on compatible periodic lattices. The direct overdamped gravity interpretation fails universality because its response depends on probe diffusivity. A separate inertial calibration produces $\mathbf{a} = -\text{grad } \Phi$ only by importing $\mathbf{u} = -\Phi/c_*^2$ and postulating the cancelling response coefficient.

The current contribution is therefore a bounded classical stochastic-model program, not a completed quantum theory of gravity. It supplies explicit theorems, a non-derivation result, an internal model-selection slope, a reproducible computational comparison, and precise failure conditions. Source dynamics, independent measurement of temporal opportunity, matter and light propagation, covariance, conservation, backreaction, quantum structure, and a prediction beyond general relativity remain open. Those limits preserve the original QGTCD innovation as a testable research hypothesis while preventing the hypothesis from being mistaken for a result already established.

Data and code availability

The private reproducibility package contains the endpoint-kernel script and frozen results, the three-kernel comparison script and results, the periodic stationary-limit script and results, versioned claim ledgers, and the dated source map. Specifically, `model/qgtcd_draft2_kernel_comparison.py` generates `model/draft2-kernel-comparison-results.json`, and `model/qgtcd_draft6_stationary_limit.py` generates `model/draft6-stationary-limit-results.json`. The current review PDF is generated from this exact Markdown manuscript. Public release, if later authorized, must use independently reviewed files and a newly frozen SHA-256 manifest.

Authorial provenance and research chronology

Artificial intelligence was not involved in authoring or originating the core ideas, scientific theories, hypotheses, or source-based research presented in this paper. The underlying research program was not written or developed by AI. This body of work is Micah Blumberg's original research, developed through a continuous program spanning 2011-2026. Its central conceptual foundations were developed and recorded before ChatGPT's public release on November 30, 2022.

In later manuscript development, AI tools were used only in bounded supporting roles, including assistance with mathematical and computational formulation, code, grammar and spelling, and checks against medical, biological, and physics literature. These tools did not supply the original theories or authorship. Micah Blumberg selected the claims and sources, directed the analysis, approved the final language, and remains responsible for the paper's arguments, interpretations, and errors.

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